



7.6.5 Wrap-Up: Ticket Out the Door

1. $\triangle FHX$ has angle measures $m\angle HXF = 90^\circ$ and $m\angle HFX = 30^\circ$, and side length $HX = 6\sqrt{3}$. What are the exact measures of $\overline{FX}$ and $\overline{FH}$?



Unit 7, Lesson 7: The Unit Circle and its Coordinates



Benchmarks

MA.912.T.1.1 Define trigonometric ratios for acute angles in right triangles.

MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean Theorem.



Learning Targets

- ☐ I can explore relationships between the coordinates of vertices when the triangles are in different quadrants.
- ☐ I can use side ratios of 45 – 45 – 90 triangles to solve problems.
- ☐ I can use side ratios of 30 – 60 – 90 triangles to solve problems.



7.7.1 Warm-Up: Between Two Numbers



1. A Florida Box Turtle's average speed is about 0.17 miles per hour.

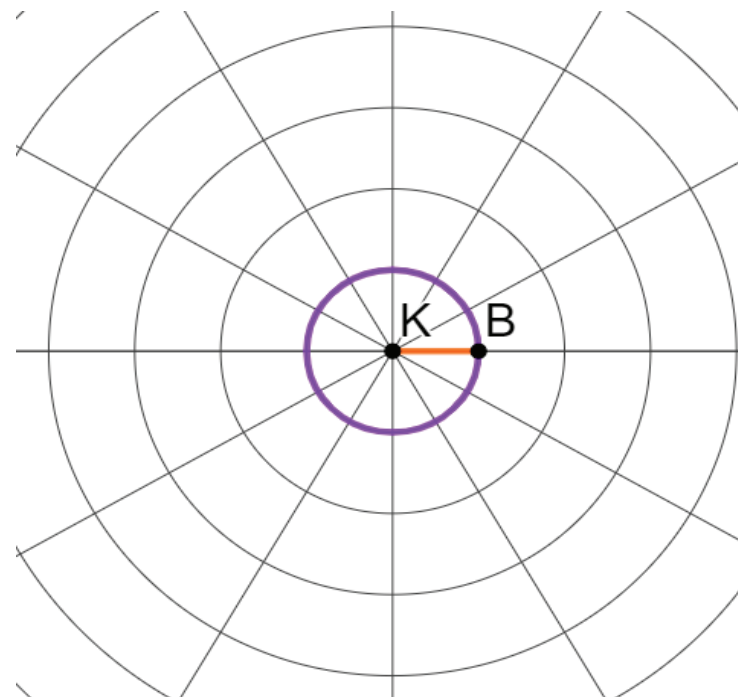
The North American X-15's top speed is about 163 miles per hour.

How much time would it take for each to travel across Florida, a distance of about 500 miles?



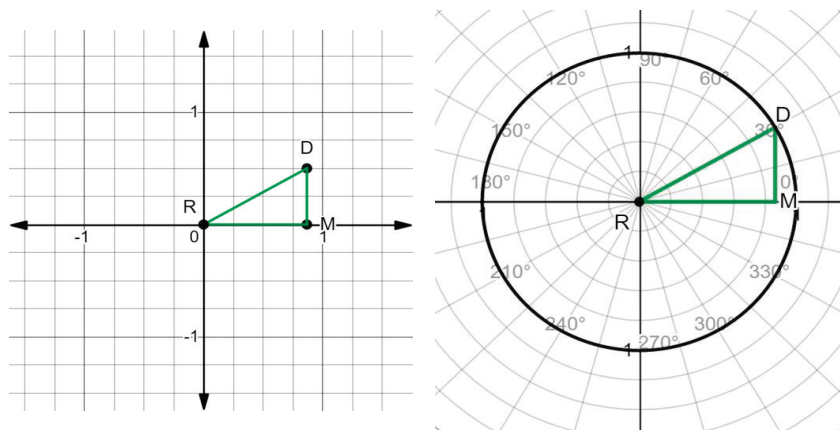
7.7.2 Exploration: The Unit Circle

1. Circle K is shown on polar graph paper, where the length of $\overline{KB}$ is 1 unit.



- a. Draw a circle that has a radius of 2 units.
- b. Draw a circle that has a radius of 3 units.

2. $\triangle RDM$ is a right triangle formed by drawing a 30° angle at the origin. $\triangle RDM$ is shown on both a coordinate grid and on a polar graph, where $m\angle MRD = 30^\circ$, $m\angle RDM = 60^\circ$, $m\angle DMR = 90^\circ$, $RD = 1$, $DM = \frac{1}{2}$, and $RM = \frac{\sqrt{3}}{2}$.



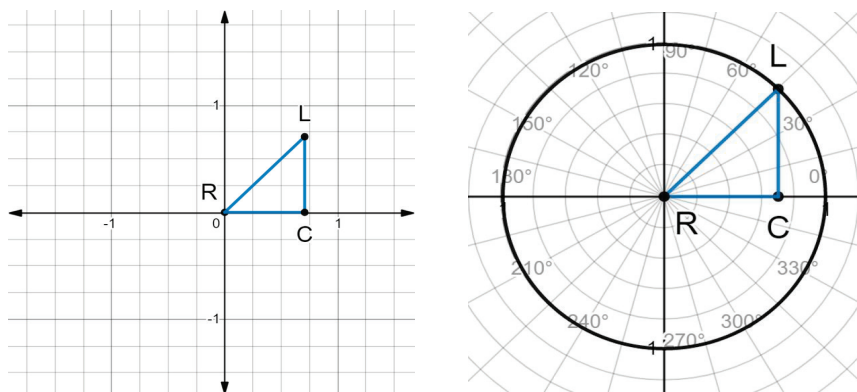
- a. Discuss with your partner any similarities and differences between the graphs. Summarize your discussion.
- b. Work with your partner to determine the coordinates of point D .

- c. Work with your partner to complete the table. Leave answers in exact form.

Angle	Cosine	Sine	Tangent
30° ($m\angle R$)			

- d. Discuss with your partner how the coordinates of point D can be found using the values from the table. Make a conjecture about the relationship between the values of cosine and sine, and the coordinate pair formed by the angle.

3. $\triangle RLC$ is a right triangle formed by drawing a 45° angle at the origin. $\triangle RLC$ is shown on both a coordinate grid and on a polar graph, where $m\angle CRL = 45^\circ$, $m\angle RLC = 45^\circ$, $m\angle RCL = 90^\circ$, $RL = 1$, $RC = \frac{\sqrt{2}}{2}$, and $LC = \frac{\sqrt{2}}{2}$.

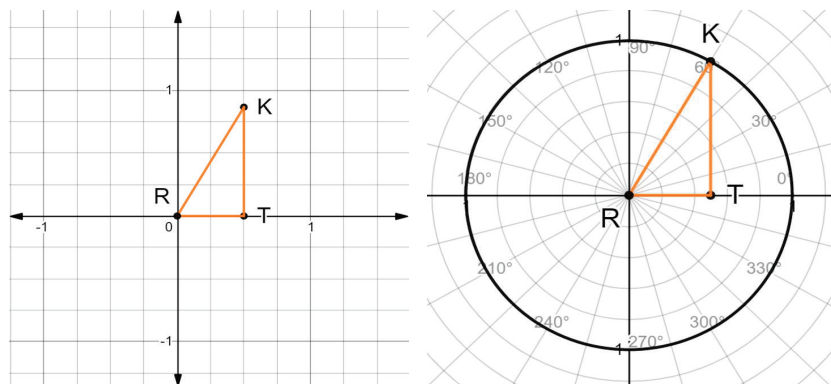


- a. Determine the coordinates of point L .
- b. Work with your partner to complete the table. Leave answers in exact form.

Angle	Sine	Cosine	Tangent
45° ($m\angle R$)			

- c. Discuss with your partner whether or not the conjecture that was made about the coordinates of point D could be applied to determine the coordinates of point L . If necessary, revise your conjecture.

4. $\triangle RKT$ is a right triangle formed by drawing a 60° angle at the origin. $\triangle RKT$ is shown on both a coordinate grid and on a polar graph, where $m\angle TRK = 60^\circ$, $m\angle RKT = 30^\circ$, $m\angle RTK = 90^\circ$, $RK = 1$, $RT = \frac{1}{2}$, and $TK = \frac{\sqrt{3}}{2}$.



a. Determine the coordinates of point K .

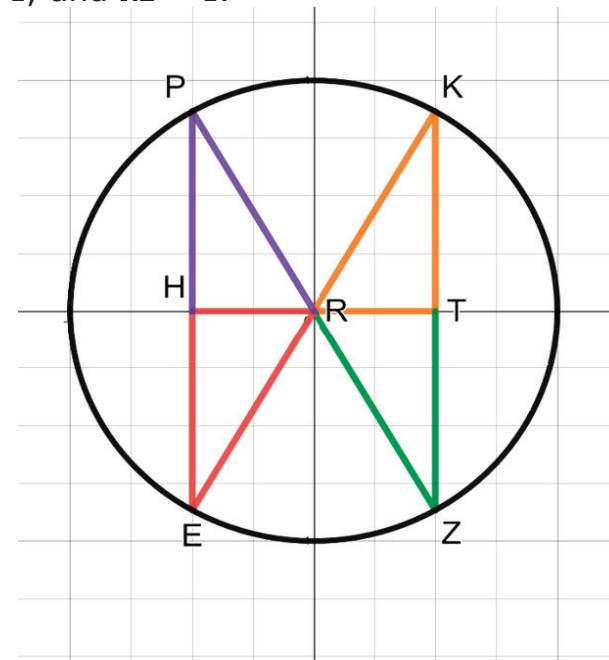
- b. Work with your partner to complete the table. Leave answers in exact form.

Angle	Sine	Cosine	Tangent
60° ($m\angle R$)			

- c. Discuss with your partner how the coordinates will change if the triangle is in a different quadrant. Make a prediction.



5. Four triangles were formed by drawing angles of 60° from the origin. $m\angle TRK = 60^\circ$, $m\angle PRH = 60^\circ$, $m\angle HRE = 60^\circ$, and $m\angle TRZ = 60^\circ$. $RK = 1$, $RP = 1$, $RE = 1$, and $RZ = 1$.

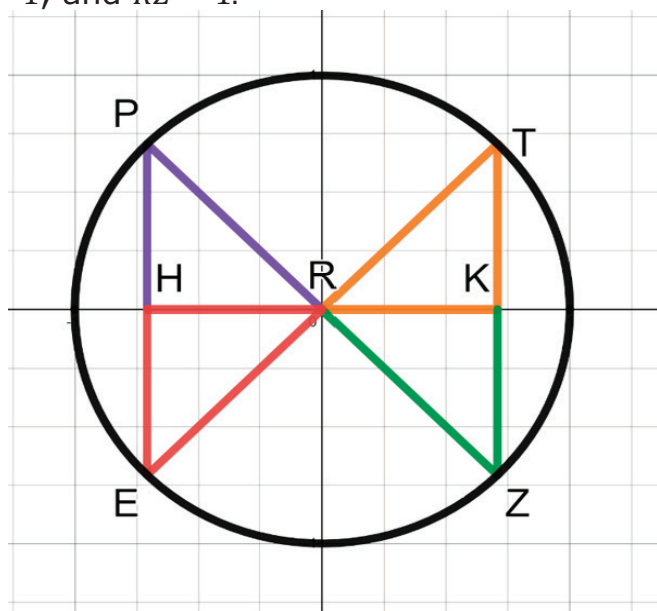


a. Complete the table.

Point	Coordinates
K	$\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$
P	
E	
Z	

- b. Points K , P , E , and Z lie on circle R . What is the radius of circle R ?
- c. What type of special triangles are $\triangle RKT$, $\triangle RPH$, $\triangle REH$, and $\triangle RZT$?

6. Four triangles were formed by drawing angles of 45° from the origin. $m\angle TRK = 45^\circ$, $m\angle PRH = 45^\circ$, $m\angle HRE = 45^\circ$, and $m\angle KRZ = 45^\circ$. $RT = 1$, $RP = 1$, $RE = 1$, and $RZ = 1$.

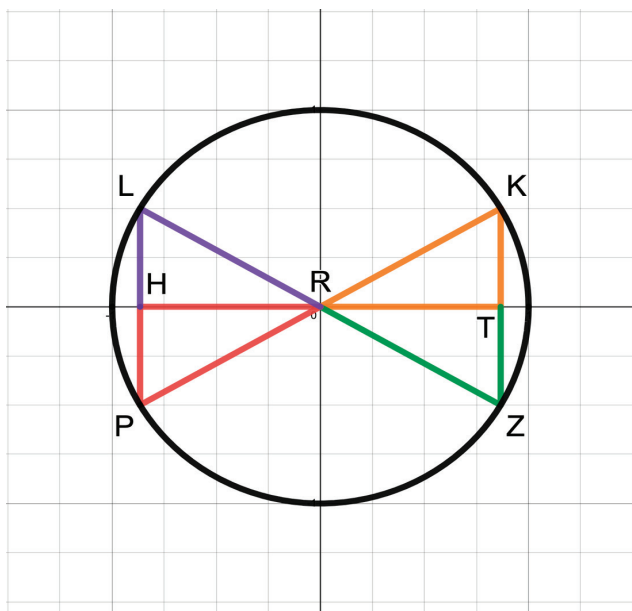


- a. Complete the table.

Point	Coordinates
T	$\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$
P	
E	
Z	

- b. Points T , P , E , and Z lie on circle R . What is the radius of circle R ?
- c. What type of special triangles are $\triangle RTK$, $\triangle RPH$, $\triangle REH$, and $\triangle RKZ$?

7. Four triangles were formed by drawing angles of 30° from the origin. $m\angle TRK = 30^\circ$, $m\angle PRH = 30^\circ$, $m\angle HRL = 30^\circ$, and $m\angle TRZ = 30^\circ$. $RK = 1$, $RP = 1$, $RL = 1$, and $RZ = 1$.



- a. Complete the table.

Point	Coordinates
K	$\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$
L	
P	
Z	

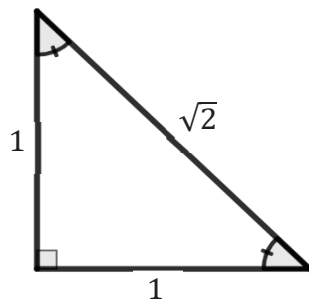
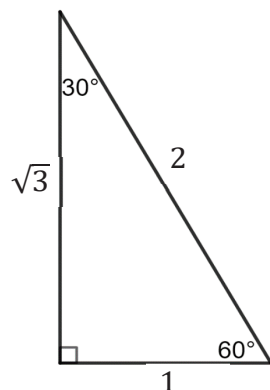
- b. Points K , P , L , and Z lie on circle R . What is the radius of circle R ?
- c. What type of special triangles are $\triangle RKT$, $\triangle RPH$, $\triangle RLH$, and $\triangle RZT$?

8. For each type of special triangle, what similarities do the coordinates for each vertex located on the circle share with one another?
9. For each type of special triangle, what differences do the coordinates for each vertex located on the circle share with one another?



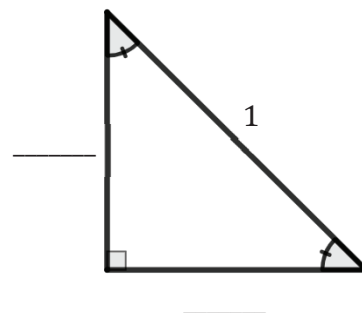
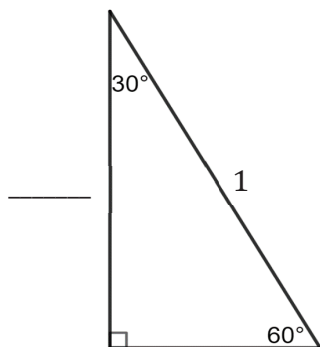
7.7.3 Bring It Together: The Unit Circle

In the Exploration, all of the triangles were one of the two special triangles, $30 - 60 - 90$ or $45 - 45 - 90$. The relationships of each of the side lengths for these triangles are shown.

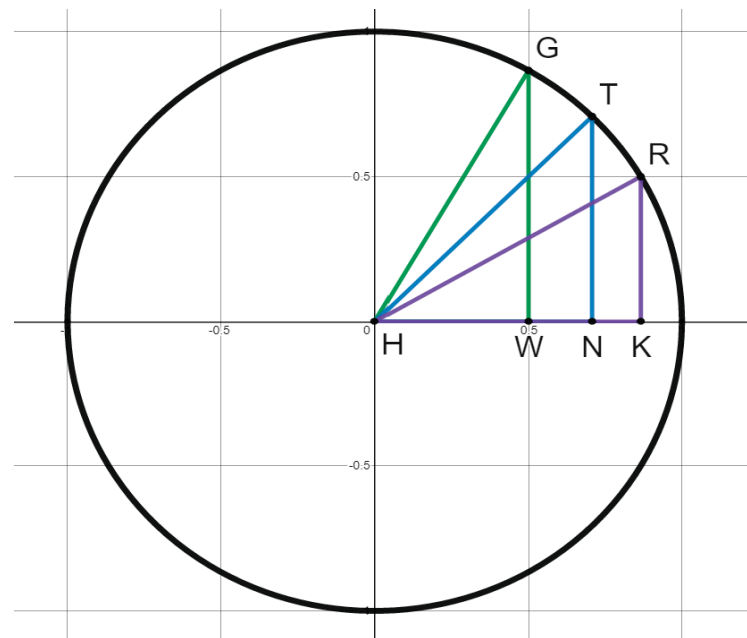


In the circle, the hypotenuse of the triangle was the radius of the circle, which was 1. When a circle has a radius of 1 and is centered at the origin, it is called the unit circle. In the diagrams of the two special triangles the hypotenuse is not 1.

- Fill in the blanks to show the relationships of the special triangles when the hypotenuse is 1.



- A unit circle with $\triangle HKR$, $\triangle HGW$, and $\triangle HNT$ is shown, where $m\angle KHR = 30^\circ$, $m\angle NHT = 45^\circ$, $m\angle WHG = 60^\circ$, $m\angle GWH = 90^\circ$, $m\angle HNT = 90^\circ$, and $m\angle HKR = 90^\circ$.



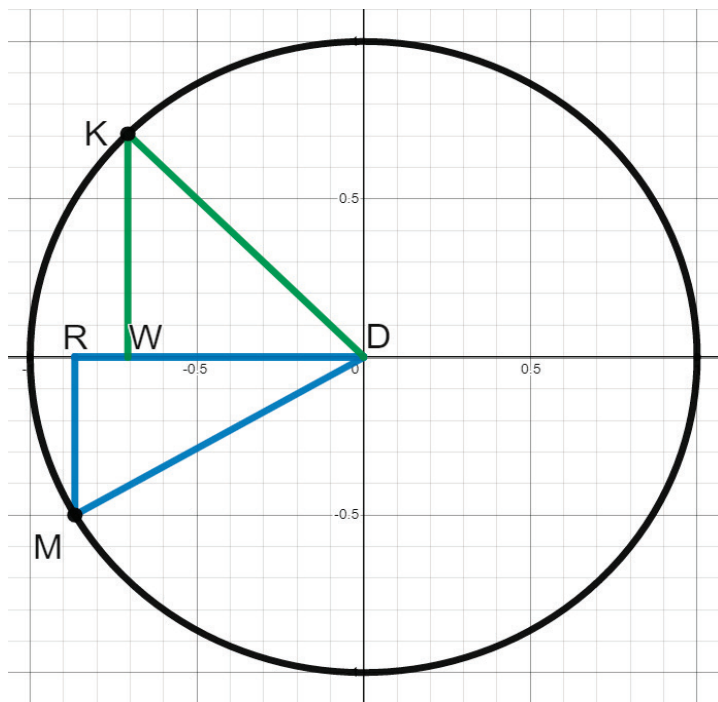
Complete the table.

Point	Coordinates
G	
R	
T	



7.7.4 Try It: Finding Coordinates in the Unit Circle

1. A unit circle is shown where $\triangle KDW$ and $\triangle MRD$ are right triangles, $m\angle KDW = 45^\circ$, and $m\angle RDM = 30^\circ$.



Complete the table.

Point	Coordinates
K	
M	



7.7.5 Wrap-Up: Reflection

1. Complete the table.

The terms and information used in the Exploration that I knew were ...	Something I wondered during the Exploration was ...	Something I learned during the Exploration was ...